

G8-B-PROFILE-MECHANISM-AUDIT-v1: Does rescaled profile force $\alpha > 0$ with weight?

ZFL-CX

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1 Frozen state

$G7 = FROZEN, G8-01 = G8-02 = G8-03 = KILLED$	$G8-DESIGN-AUDIT-v2 = DONE$ (571b5b0)
$A = BLOCKED, C = BLOCKED$ (E-07/E-08 D_{te_i} barrier)	$B = OPEN$ for mechanism audit only, NO

2 Candidate B definition

Rescaling: $u^{(n)}(y, s) = r_n u(r_n y, T^* + r_n^2 s)$, $r_n \rightarrow 0$ concentration scale from $C_F^{conc} + C_I^{freq}$. Assume $u^{(n)} \rightarrow \bar{u}$ limit (ancient solution to NS/Euler), compatible with $\mathcal{C}_{PROVEN}$. $\bar{\omega} = \nabla \times \bar{u}$, $\bar{S} = \nabla_{sym} \bar{u}$, $\bar{\xi} = \bar{\omega}/|\bar{\omega}|$, $\bar{\alpha} = \bar{\xi}^T \bar{S} \bar{\xi}$. $W_+^{(n)} \sim r_n^{-1} \int |\bar{\omega}|^2 \bar{\alpha}_+$ (scaling).

3 Falsifiable question

Can every blow-up profile compatible with $C_F^{conc} + C_I^{freq} + C_J^{neg}$ have $\bar{\alpha} \leq 0$? If NO: there exists identity/law of profile forcing region $\{\bar{\alpha} > 0\}$ with sufficient measure to affect W_+ .

We need NOT pointwise, but: $\int_{\{\bar{\alpha} > 0\}} |\bar{\omega}|^2 \bar{\alpha} \geq c_0 > 0$ with c_0 independent of n , and temporal weight $\int^{T^*} W_+(t) dt = +\infty$ forced.

4 Barriers - strong form

1. $\boxed{\text{nontrivial profile} \not\Rightarrow \alpha > 0}$. Need stronger.
2. $\boxed{\alpha > 0 \text{ at a point} \not\Rightarrow \int W_+ dt < \infty}$. Need measure/scale+time.
3. No assumption $\xi^T S \xi > 0$, that is to be proved. C_J^{neg} allows compressive.
4. No confusion concentration $\not\Rightarrow$ stretching.
5. Must avoid eigenframe D_{te_i} formula, else falls back to C barrier.
6. No dimensional $\|u\|_3 \|\nabla \omega\|_2^2$ as bridge.
7. Only identity derived from limit equations (NS/Euler + incompressibility + $\mathcal{C}_{PROVEN}$ constraints).

5 What would count as bridge?

Type 1 - Liouville: $\bar{u}$ nontrivial ancient + $C_J^{neg} \Rightarrow \int |\bar{\omega}|^2 \bar{\alpha} > 0$ forced by enstrophy balance.
Type 2 - Geometric: divergence-free forces $\text{tr } \bar{S} = 0$, but combined with concentration C_F^{conc} forces

eigenvalue $\lambda_1 > 0$ dominating in region where $\bar{\omega}$ concentrates. Type 3 - Temporal: profile law gives r_n^{-1} scaling $\Rightarrow \int_0^{T^*} W_+ \sim \sum r_n^{-1} \rightarrow \infty$ unless $c_0 = 0$, so $c_0 > 0$ would contradict integrability? Actually need reverse: show $\int W_+ < \infty$ from profile, or $\perp$.

All require new identity, not just compactness.

6 Kill criteria for B-audit

KILLED if: - only proves $\bar{u} \neq 0$ without sign, - only proves $\bar{\alpha} > 0$ at isolated point without weight, - uses $D_t e_i$ or $\|S\|_\infty$, - reduces to $\|S\|_{L^3}^3$ estimate, - assumes smallness not derived from $T^* < \infty$, - no quantitative $\int |\bar{\omega}|^2 \bar{\alpha} \geq c_0$.

7 Decision rule

If no such identity found:

$G8-DESIGN-v2 = CLOSED/BLOCKED$

$G8 = 01 + 02 + 03 \text{ KILLED, } A/B/C \text{ E}$
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Then whole conceptual class scale/profile/ S - Ω exhausted, ZFL clean, no fabrication. If identity found: it becomes sole authorized G8-04 with explicit mechanism, target, kill criteria.

8 Status

This audit is NOT G8-04. Only mechanism check for B.
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